

Rian Edwards

+614 1160 3291 | edwardsrian8@gmail.com | [linkedin.com/in/rian-edwards-668641271](https://www.linkedin.com/in/rian-edwards-668641271)

Career Objective

Motivated fourth-year Mechatronics Engineering student at the University of Technology Sydney, with a strong interest in automation, robotics, sensing systems, CAD, machine learning, and embedded electronics. Experienced in hands-on prototyping and technical problem-solving, and eager to apply practical engineering expertise in fast-paced, real-world environments.

Technical Skills

Programming: C/C++, Java, Python, ROS 2, Arduino, MATLAB, Java, JavaScript, HTML, CSS, Git/GitHub

Machine Learning: YOLOv8 OBB, scikit-learn, dataset preparation, model training and evaluation

Design & CAD: SolidWorks, 3D printing, Mechanical Assembly

Embedded Systems & Electronics: PCB design, Soldering, Motor Drivers, Sensor Integration, M5Stack

Software: Adobe (Premier Pro, Photoshop), Azure, Microsoft Office (Word, Excel, PowerPoint), VS Code

Education

Bachelor of Engineering (Honours), Mechatronics

February 2023 – December 2026

University of Technology Sydney (UTS)

WAM:80.23

Work Experience

Winter Scholarship - IntelliForce Hydraulic/Pneumatic Safety System

June 2026 - Ongoing

UTS Robotics Institute

- Developing a ESP32 based hydraulic and pneumatic pump safety system for a mining industrial partner
- Designed and validated custom PCB for switching solenoid valve
- Developed and debugged embedded firmware (C++) for ESP32 and M5Stack platforms, including runtime-configurable safety parameters

Intern - Plant Development & Engineering

December 2025 – February 2026

PT Saptaindra Sejati (Subsidiary of Adaro Energy)

- Led feasibility study for collaborative robotic welding system eliminating high-risk manual welding at heights up to 4.5 m; developed TER/TOR documents and conducted vendor evaluation using a weighted decision matrix across five suppliers, resulting in successful procurement and commissioning
- Designed and simulated Arduino-based IMU alert system (TinkerCAD/Wokwi) for inclination-triggered warnings at >9 degree threshold; findings contributed to a PLC/HMI inclinometer monitoring system reducing swing bearing failures in Komatsu excavators

Intern - Software Development & Business Analysis

January 2025 – March 2025

PricewaterhouseCoopers (PwC)

- Contributed to the development of a web application using Python (Django), JavaScript, and CSS, focusing on feature implementation and debugging across development stages.
- Collaborated with the development team to deliver functional and user-friendly app modules in an agile environment.
- Authored detailed Technical Specification Documents (TSD) and Functional Specification Documents (FSD) using Excel to define app requirements and streamline communication between technical and non-technical stakeholders.
- Designed and executed test scripts, including stress tests, to ensure the application's performance, reliability, and stability.

Relevant Projects

Sort-O: Autonomous Plastic Sorting Robot (Portfolio: jun-je0ng.github.io/Robotics-Studio-2)

- Led the Machine Learning subsystem within a 4-person team building a fully autonomous robot that detects, classifies and sorts plastic waste using a UR3e robotic arm, depth camera and ROS2.
- Fine-tuned YOLOv8 OBB model on a custom dataset achieving mAP@50 of 0.991, precision 0.988 and recall 0.991 across all three classes; model outputs rotation angle enabling gripper alignment with bottle orientation.
- Designed and implemented a K-Means clustering algorithm to optimise robot pick order by spatial proximity, reducing unnecessary arm movement and improving cycle efficiency.
- Collaborated across perception, motion control and execution subsystems, defining shared data interfaces and message formats that enabled parallel development across the full team
- Presented project progress and results to a client and academic panel through structured demonstrations and technical documentation

Pedometer Tracker

- Designed and implemented an accelerometer-based step detection algorithm using the ADXL335 sensor to track walking, running, and stationary states with 90%+ accuracy.
- Led embedded software subsystem design, with user-guided positioning enabling step counting and pace identification to display a calorie burn estimator.
- Assisted electronics subsystem design, including analogue filters and PCB schematic design layout using Altium, enabling compact and power-efficient tracking.
- Collaborated with a four-member interdisciplinary team to integrate electronics, mechanical casing, and user interface within 12-week project timeframe.

Robot Pick and Place

- Designed a MATLAB-based algorithm for detecting and classifying objects by analysing shape and colour, using RGB values, centroid positions, and circularity metrics.
- Applied key image processing techniques such as greyscale transformation, thresholding, and contour analysis-to extract distinguishing object features.
- Utilised camera calibration to transform pixel coordinates into spatial coordinates, allowing accurate real-world positioning for robotic manipulation.
- Executed pick-and-place tasks with the DoBot robot by integrating robotic kinematics principles for precise arm movement and control.

Warman Challenge 2024

- Integrated mechanical and electronic subsystems to develop a functional autonomous robot capable of completing a complex delivery task under competition conditions.
- Coordinated interaction between actuators, sensors, and motors to ensure reliable system behaviour and precise mechanical responses.
- Conducted system-level testing and troubleshooting to resolve issues in mechanical-electrical alignment, improving consistency and task success rate.